

Section 3.7 Exercises

Directions: For exercises 1 - 4, find the domain of each function. State your answer in interval notation.

1. $f(x) = \frac{x-1}{x+2}$

2. $f(x) = \frac{x+1}{x^2-1}$

3. $f(x) = \frac{x^2+4}{x^2-2x-8}$

4. $f(x) = \frac{x^2+4x-3}{x^4-5x^2+4}$

Directions: For exercises 5 - 14, find the domain, vertical asymptotes, and horizontal asymptotes of each function.

5. $f(x) = \frac{4}{x-1}$

6. $f(x) = \frac{2}{5x+2}$

7. $f(x) = \frac{x}{x^2-9}$

8. $f(x) = \frac{x}{x^2+5x-36}$

9. $f(x) = \frac{3+x}{x^3-27}$

10. $f(x) = \frac{3x-4}{x^3-16x}$

11. $f(x) = \frac{x^2-1}{x^3+9x^2+14x}$

12. $f(x) = \frac{x+5}{x^2-25}$

13. $f(x) = \frac{x-4}{x-6}$

14. $f(x) = \frac{4-2x}{3x-1}$

Directions: For exercises 15 - 19, find the x - and y -intercepts for each function.

15. $f(x) = \frac{x+5}{x^2+4}$

16. $f(x) = \frac{x}{x^2-x}$

17. $f(x) = \frac{x^2+8x+7}{x^2+11x+30}$

18. $f(x) = \frac{x^2+x+6}{x^2-10x+24}$

19. $f(x) = \frac{94-2x^2}{3x^2-12}$

Directions: For exercises 20 - 24, find the slant asymptote of each function.

$$20. f(x) = \frac{24x^2+6x}{2x+1}$$

$$21. f(x) = \frac{4x^2-10}{2x-4}$$

$$22. f(x) = \frac{81x^2-18}{3x-2}$$

$$23. f(x) = \frac{6x^3-5x}{3x^2+4}$$

$$24. f(x) = \frac{x^2+5x+4}{x-1}$$

Directions: For exercises 25 - 34, find the intercepts, the vertical asymptotes, and the horizontal or slant asymptote of each function. Use that information to sketch the graph.

$$25. p(x) = \frac{2x-3}{x+4}$$

$$26. q(x) = \frac{x-5}{3x-1}$$

$$27. s(x) = \frac{4}{(x-2)^2}$$

$$28. r(x) = \frac{5}{(x+1)^2}$$

$$29. f(x) = \frac{3x^2-14x-5}{3x^2+8x-16}$$

$$30. g(x) = \frac{2x^2+7x-15}{3x^2-14x+15}$$

$$31. a(x) = \frac{x^2+2x-3}{x^2-1}$$

$$32. b(x) = \frac{x^2-x-6}{x^2-4}$$

$$33. h(x) = \frac{2x^2+x-1}{x-4}$$

$$34. k(x) = \frac{2x^2-3x-20}{x-5}$$